TAYYABA ZUBAID

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Tayyaba Zubaid

SUMMARY

Computer Science undergraduate at Qarshi University with project-based experience in developing responsive web applications, interactive dashboards, and algorithm-based systems. Proficient in HTML, CSS, JavaScript, C++, and Java, with practical application of object-oriented programming, data structures, shortest-path algorithms, and real-time data processing. Focused on applying structured problem-solving and clean interface design to develop functional, user-oriented software solutions.

TECHNICAL SKILLS

Programming Languages:
C++, Java, Python, JavaScript, SQL

Web Technologies:
HTML5, CSS3, Responsive Web Design

Frameworks & Libraries:
Java Swing, Bootstrap

Data Structures & Algorithms:
Arrays, Linked Lists, Trees, Graphs, Dijkstra's Algorithm, Segment Tree

Object-Oriented Programming:
OOP principles, Design Patterns (Observer)

Tools & Platforms:
Git, GitHub, VS Code

PROJECTS

Deenify - An Islamic Website Design

- **Developed a mobile-responsive web application** using HTML5 and CSS3, focusing on clean layouts, proper spacing, and a mobile-first user interface.
- **Implemented multi-page navigation** by linking modular pages (Home, About, Features, FAQs, Contact) to improve usability and maintainability.
- **Organized and presented feature-based content** in a structured, user-friendly design to enhance readability and accessibility on small screens.

FinTrack - An Investment Dashboard

- **Developed a comprehensive financial dashboard** using Java Swing, engineering an interactive UI with dynamic graphs, custom table renderers, and a modular navigation system to track and visualize market data, investment portfolios, and profit/loss calculations.
- **Implemented core application logic** using Object-Oriented principles, including an observer design pattern to ensure real-time data synchronization between UI panels and building a robust profit/loss calculator with multi-currency support.

Pokemon Info Card - A Pokemon Encyclopedia

- **Developed a dynamic Pokémon encyclopedia** using JavaScript and PokeAPI, featuring real-time data fetching for 1,000+ entries and a custom-built local image repository for optimized performance.
- **Designed a responsive, themed UI/UX** with advanced navigation features, including generation-based filtering, interactive stat visualizations, and seamless pagination to enhance user engagement.
- **Engineered robust search and filtering systems** to provide an intuitive user experience, ensuring consistent branding and high-resolution visual delivery across all application modules.

Disaster Evacuation Path Planner - A Console-Based Dashboard

- **Architected a Real-Time Disaster Evacuation Engine** by developing a C++ system using Dijkstra's Algorithm and Dynamic Graphs to calculate optimal safety routes, achieving sub-10-second path recalculation during simulated urban crises.
- **Engineered High-Efficiency Risk Management** by implementing a Segment Tree data structure to process environmental hazard data (e.g., flood levels) with O(log N) efficiency, enabling instantaneous rerouting as disaster conditions evolved.
- **Developed a Modern Console-Based Dashboard** by integrating the Tabulate and Termcolor libraries to build a vibrant, color-coded UI, enhancing situational awareness by visually distinguishing between safe zones (Blue), evacuation groups (Pink), and active disaster alerts (Red).
- **Validated System Robustness via Complex Simulations** by modeling multi-point infrastructure failures and flood scenarios to ensure system reliability, aligning technical implementation with UN Sustainable Development Goals (SDG 11 & 3) for urban resilience.

Weather Dashboard - A Dynamic Weather App

- **Developed a dynamic Weather Dashboard** using HTML5, CSS3, and JavaScript, featuring real-time weather data visualization for major Pakistani cities with automated date and day synchronization.
- **Implemented adaptive UI/UX elements,** including time-based icon switching (Day/Night modes) and weather-responsive backgrounds that dynamically update based on temperature and atmospheric conditions.
- **Engineered core application logic** for seamless temperature unit conversion (Celsius/Fahrenheit) and automated hourly data refreshes to ensure accurate forecast reporting and enhanced user engagement.

Tomzora - An Image Gallery

- **Developed a dynamic image gallery application** using vanilla JavaScript and Unsplash API integration, featuring real-time search functionality across 1,000+ high-quality photos with advanced filtering by 14+ categories, dimensions (Landscape/Portrait/Square), and sorting options (Latest/Oldest/Popular).
- **Designed a fully responsive, multi-themed UI/UX** with 5 distinct color schemes (Light, Dark, Sepia, Ocean, Forest), customizable grid layouts (2-5 columns), and a mobile-optimized hamburger navigation menu that seamlessly adapts from 360px mobile devices to 1920px+ desktop displays.
- **Engineered robust state management and search persistence systems** to handle complex user interactions, including persistent favorites/downloads tracking via localStorage, dynamic view mode switching (Masonry/Uniform Grid/List View), and intelligent filter preservation across navigation changes.

EDUCATION

Intermediate — 2022 - 2024
Punjab Group of Colleges
FSc Pre-Engineering

Bachelor's in Computer Science — 2024 - Present
Qarshi University
Expected Graduation: 2028